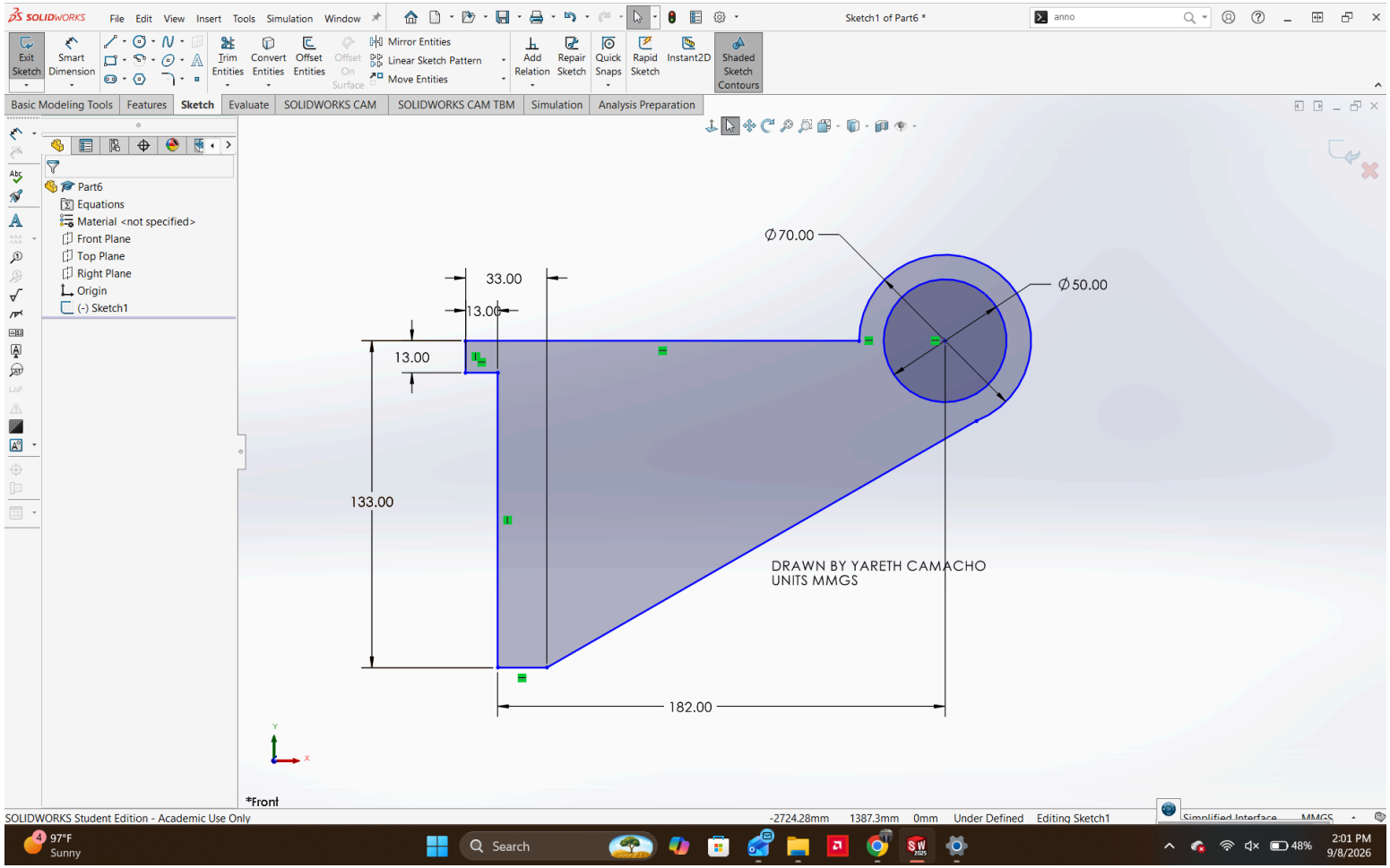
MAE-1351-001

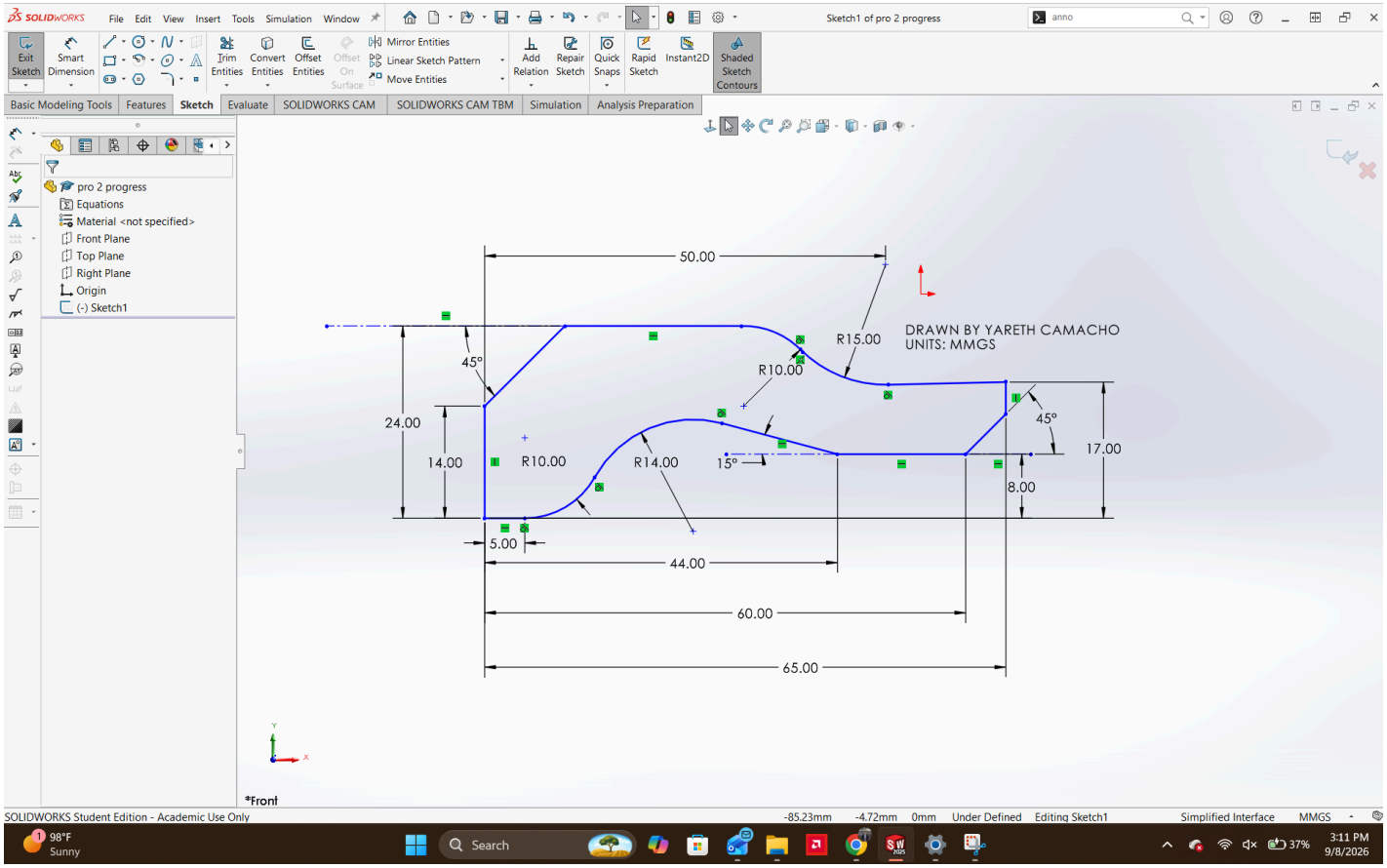
Yareth Antonio Camacho

Mechanical Engineering Design

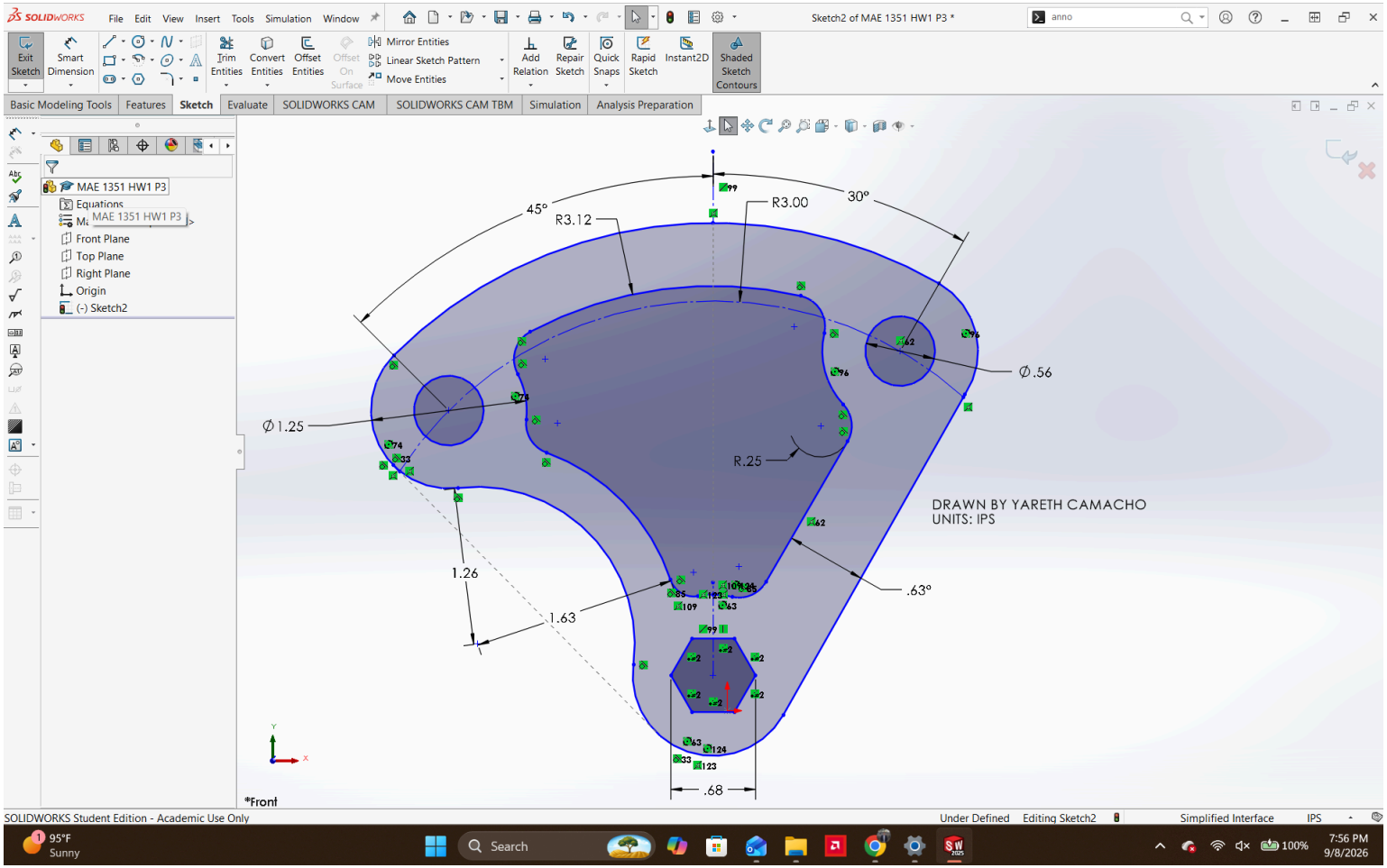
# Question 1



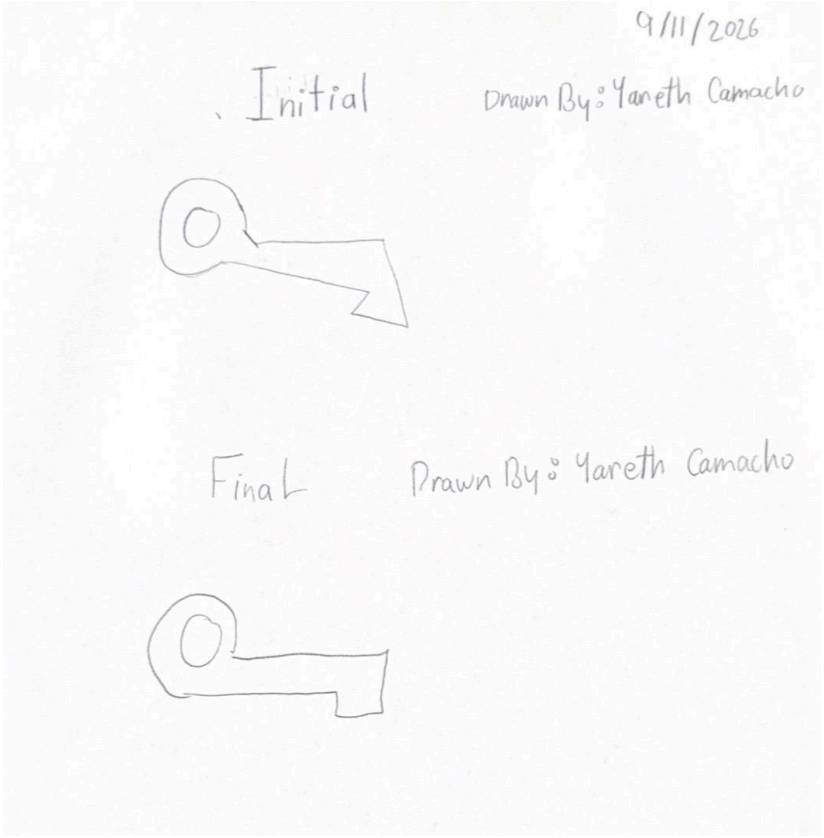
# Question 2


SOLIDWORKS
File Edit View Insert Tools Simulation Window
Sketch1 of pro 2 progress
Exit Sketch
Smart Dimension
Trim Entities
Convert Entities
Offset Entities
Mirror Entities
Linear Sketch Pattern
Move Entities
Add Relation
Repair Sketch
Quick Snaps
Rapid Sketch
Instant2D
Shaded Sketch Contours
Basic Modeling Tools
Features
Sketch
Evaluate
SOLIDWORKS CAM
SOLIDWORKS CAM TBM
Simulation
Analysis Preparation
pro 2 progress
Equations
Material <not specified>
Front Plane
Top Plane
Right Plane
Origin
(-) Sketch1
50.00
R15.00
DRAWN BY YARETH CAMACHO
UNITS: MMGS
45°
R10.00
24.00
14.00
R10.00
R14.00
15°
45°
17.00
8.00
5.00
44.00
60.00
65.00
*Front
SOLIDWORKS Student Edition - Academic Use Only
-85.23mm
-4.72mm
0mm
Under Defined
Editing Sketch1
Simplified Interface
MMGS
98°F
Sunny
Search
37%
3:11 PM
9/8/2026

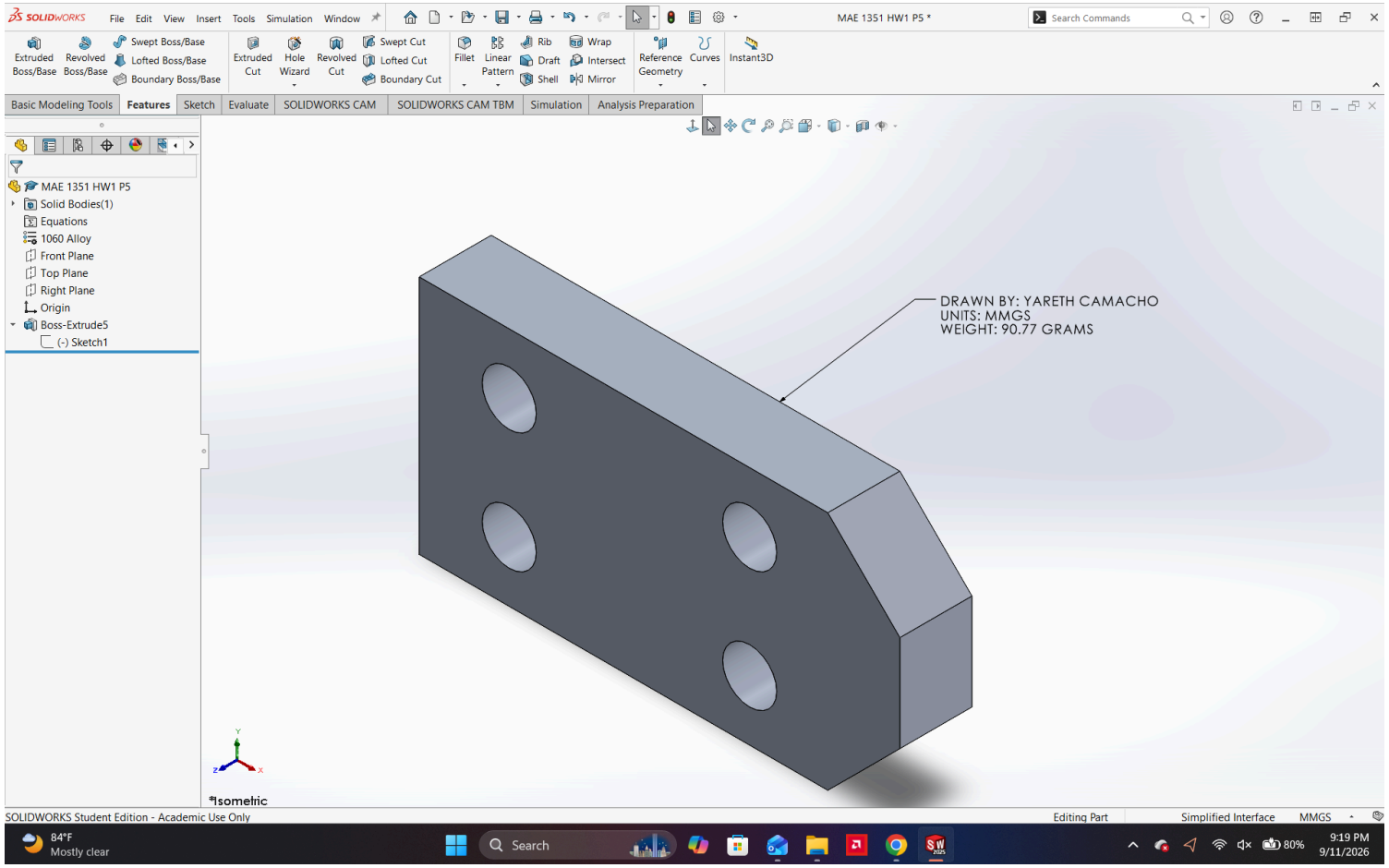
# Question 3


45°
R3.12
R3.00
30°
Ø.56
Ø1.25
R.25
DRAWN BY YARETH CAMACHO
UNITS: IPS
1.26
1.63
.63°
.68
*Front
MAE 1351 HW1 P3
Equations
Front Plane
Top Plane
Right Plane
Origin
(-) Sketch2
Sketch2 of MAE 1351 HW1 P3 *
Basic Modeling Tools
Features
Sketch
Evaluate
SOLIDWORKS CAM
SOLIDWORKS CAM TBM
Simulation
Analysis Preparation
SOLIDWORKS Student Edition - Academic Use Only
Under Defined
Editing Sketch2
Simplified Interface
IPS
95°F
Sunny
Search
7:56 PM
9/8/2026

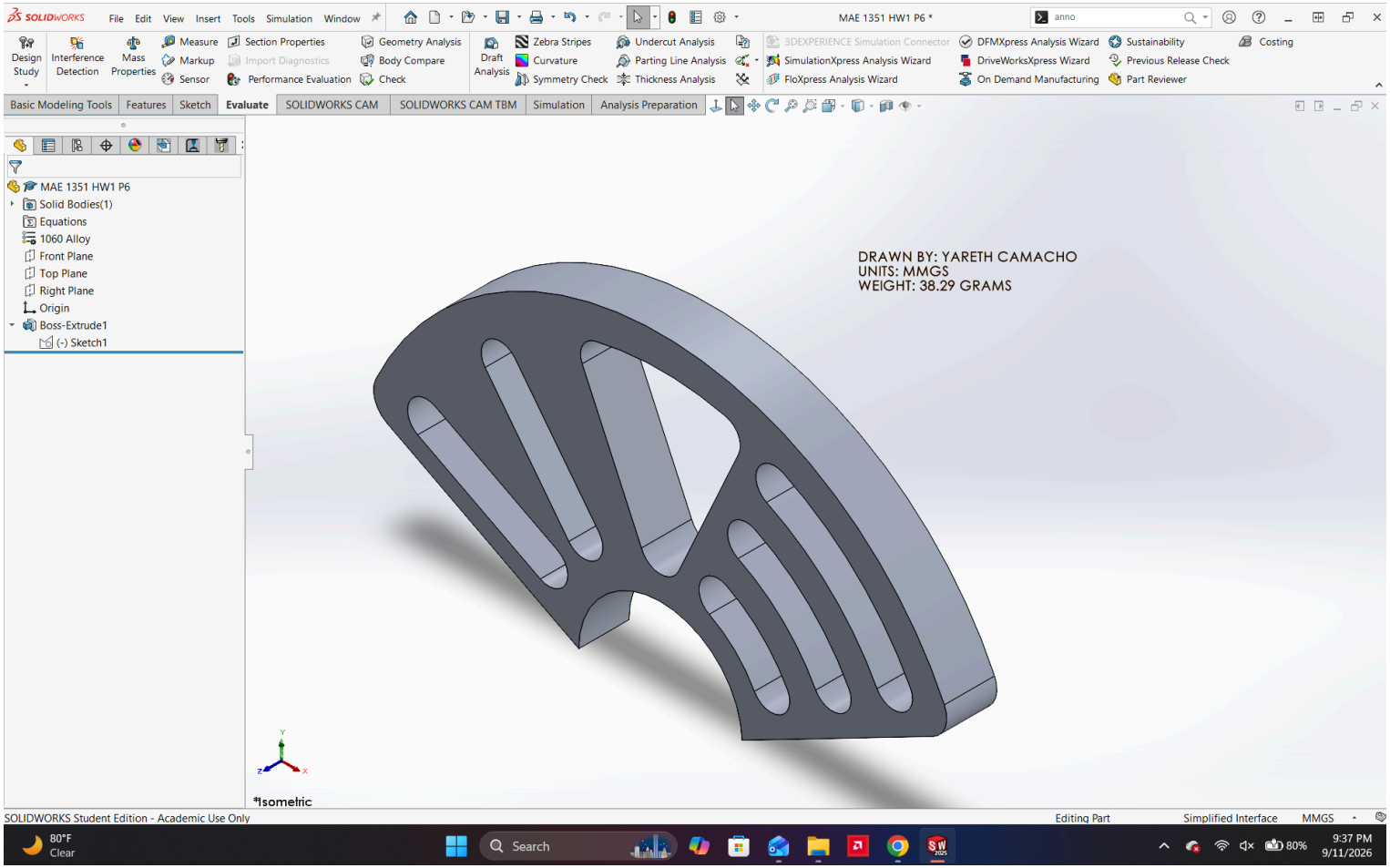
# Question 4


9/11/2026
Initial
Drawn By: Yareth Camacho
Final
Drawn By: Yareth Camacho

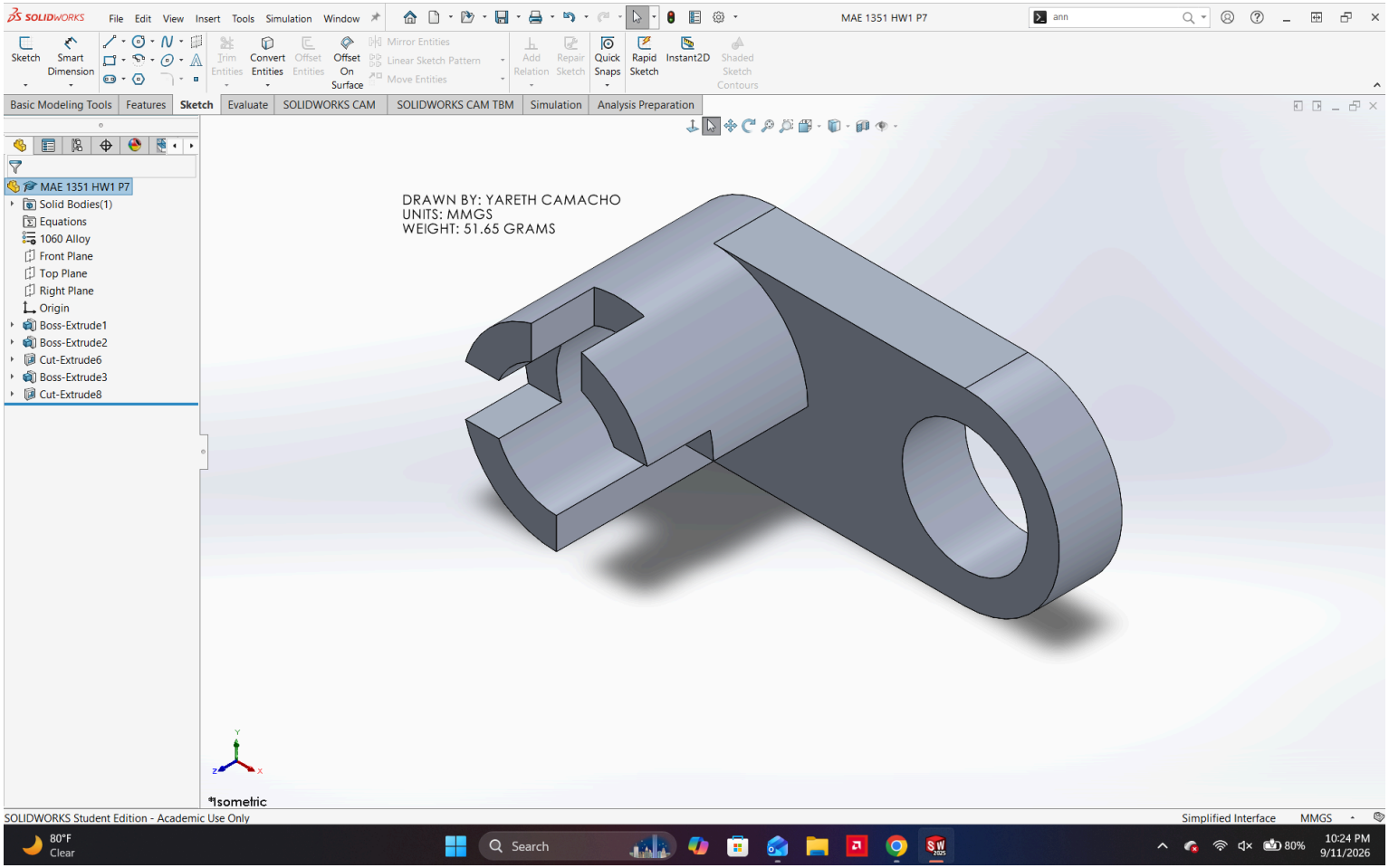
# Question 5


SOLIDWORKS
File Edit View Insert Tools Simulation Window
MAE 1351 HW1 P5 *
Search Commands
Extruded Boss/Base
Revolved Boss/Base
Swept Boss/Base
Lofted Boss/Base
Boundary Boss/Base
Extruded Cut
Hole Wizard
Revolved Cut
Swept Cut
Lofted Cut
Boundary Cut
Fillet
Linear Pattern
Rib
Draft
Shell
Wrap
Intersect
Mirror
Reference Geometry
Curves
Instant3D
Basic Modeling Tools
Features
Sketch
Evaluate
SOLIDWORKS CAM
SOLIDWORKS CAM TBM
Simulation
Analysis Preparation
MAE 1351 HW1 P5
Solid Bodies(1)
Equations
1060 Alloy
Front Plane
Top Plane
Right Plane
Origin
Boss-Extrude5
(-) Sketch1
DRAWN BY: YARETH CAMACHO
UNITS: MMGS
WEIGHT: 90.77 GRAMS
*Isometric
SOLIDWORKS Student Edition - Academic Use Only
Editing Part
Simplified Interface
MMGS
84°F
Mostly clear
Search
80%
9:19 PM
9/11/2026

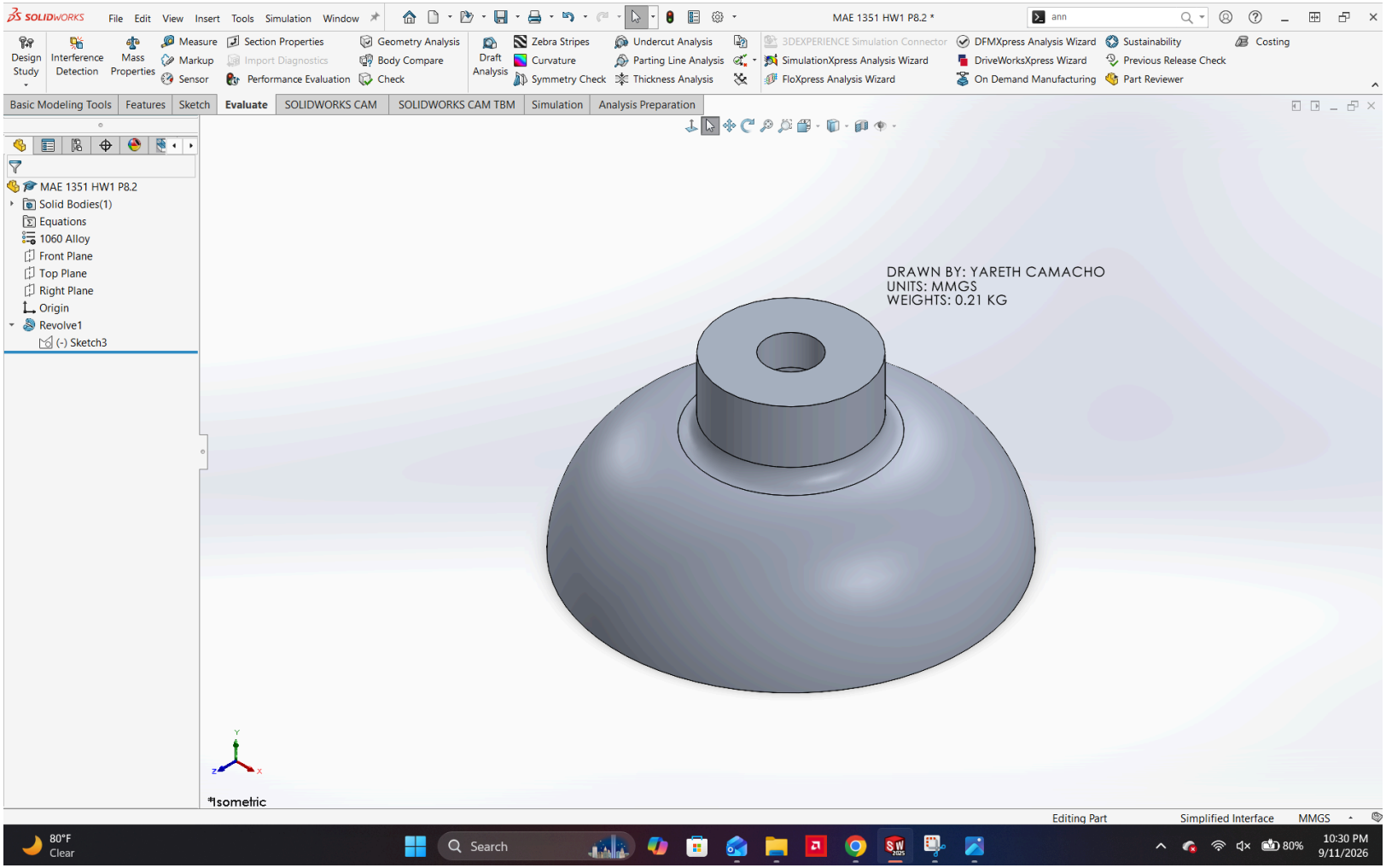
# Question 6


SOLIDWORKS
File Edit View Insert Tools Simulation Window
MAE 1351 HW1 P6 *
Design Study
Interference Detection
Mass Properties
Measure
Markup
Sensor
Section Properties
Import Diagnostics
Performance Evaluation
Geometry Analysis
Body Compare
Check
Draft Analysis
Zebra Stripes
Curvature
Symmetry Check
Undercut Analysis
Parting Line Analysis
Thickness Analysis
3DEXPERIENCE Simulation Connector
SimulationXpress Analysis Wizard
FloXpress Analysis Wizard
DFMXpress Analysis Wizard
DriveWorksXpress Wizard
On Demand Manufacturing
Sustainability
Previous Release Check
Part Reviewer
Costing
Basic Modeling Tools
Features
Sketch
Evaluate
SOLIDWORKS CAM
SOLIDWORKS CAM TBM
Simulation
Analysis Preparation
MAE 1351 HW1 P6
Solid Bodies(1)
Equations
1060 Alloy
Front Plane
Top Plane
Right Plane
Origin
Boss-Extrude1
(-) Sketch1
DRAWN BY: YARETH CAMACHO
UNITS: MMGS
WEIGHT: 38.29 GRAMS
*Isometric
SOLIDWORKS Student Edition - Academic Use Only
Editing Part
Simplified Interface
MMGS
80°F
Clear
Search
80%
9:37 PM
9/11/2026

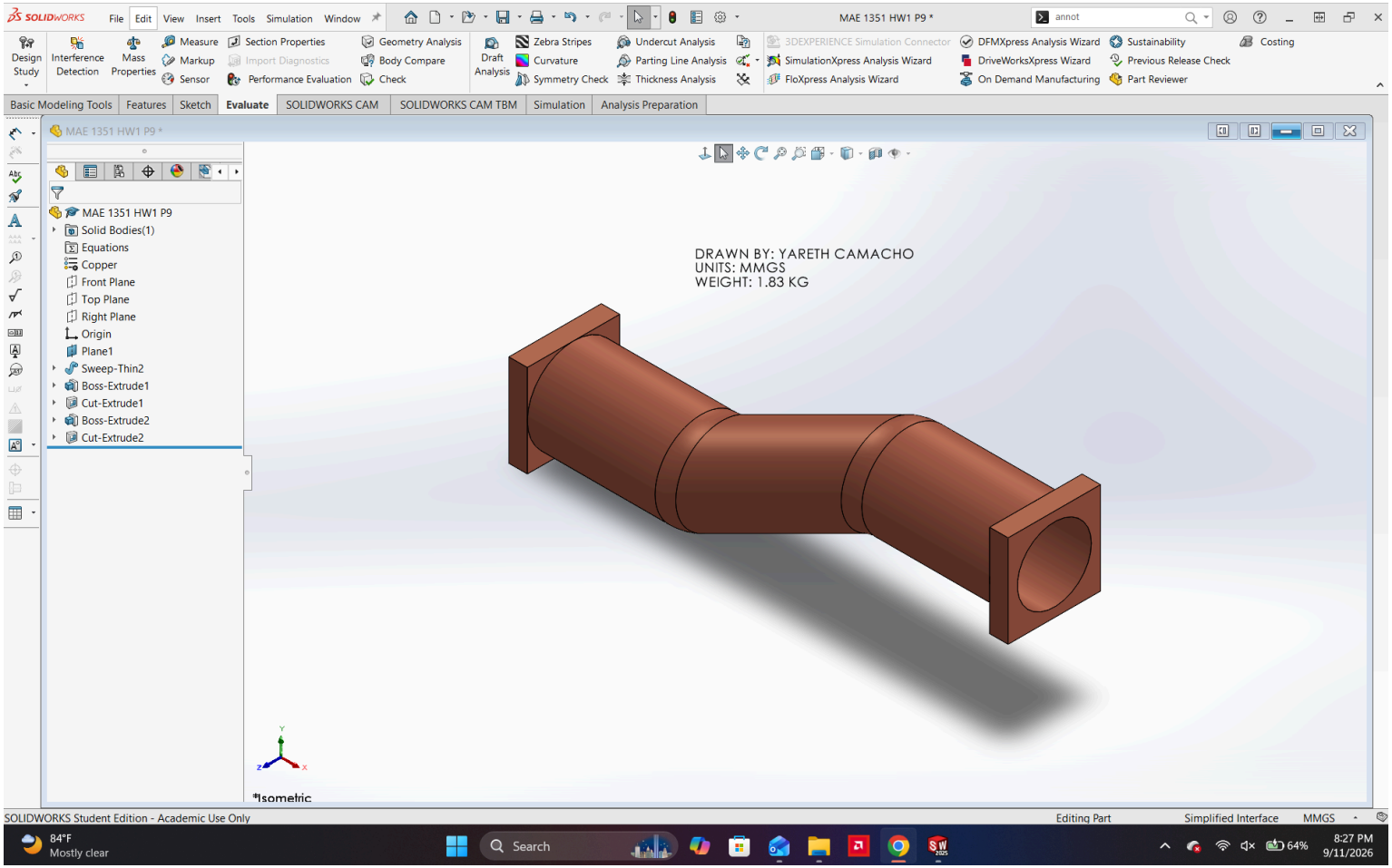
# Question 7


SOLIDWORKS
File Edit View Insert Tools Simulation Window
MAE 1351 HW1 P7
Sketch
Smart Dimension
Trim Entities
Convert Entities
Offset Entities
Offset On Surface
Mirror Entities
Linear Sketch Pattern
Move Entities
Add Relation
Repair Sketch
Quick Snaps
Rapid Sketch
Instant2D
Shaded Sketch Contours
Basic Modeling Tools
Features
Sketch
Evaluate
SOLIDWORKS CAM
SOLIDWORKS CAM TBM
Simulation
Analysis Preparation
MAE 1351 HW1 P7
Solid Bodies(1)
Equations
1060 Alloy
Front Plane
Top Plane
Right Plane
Origin
Boss-Extrude1
Boss-Extrude2
Cut-Extrude6
Boss-Extrude3
Cut-Extrude8
DRAWN BY: YARETH CAMACHO
UNITS: MMGS
WEIGHT: 51.65 GRAMS
*Isometric
SOLIDWORKS Student Edition - Academic Use Only
Simplified Interface
MMGS
80°F
Clear
Search
80%
10:24 PM
9/11/2026

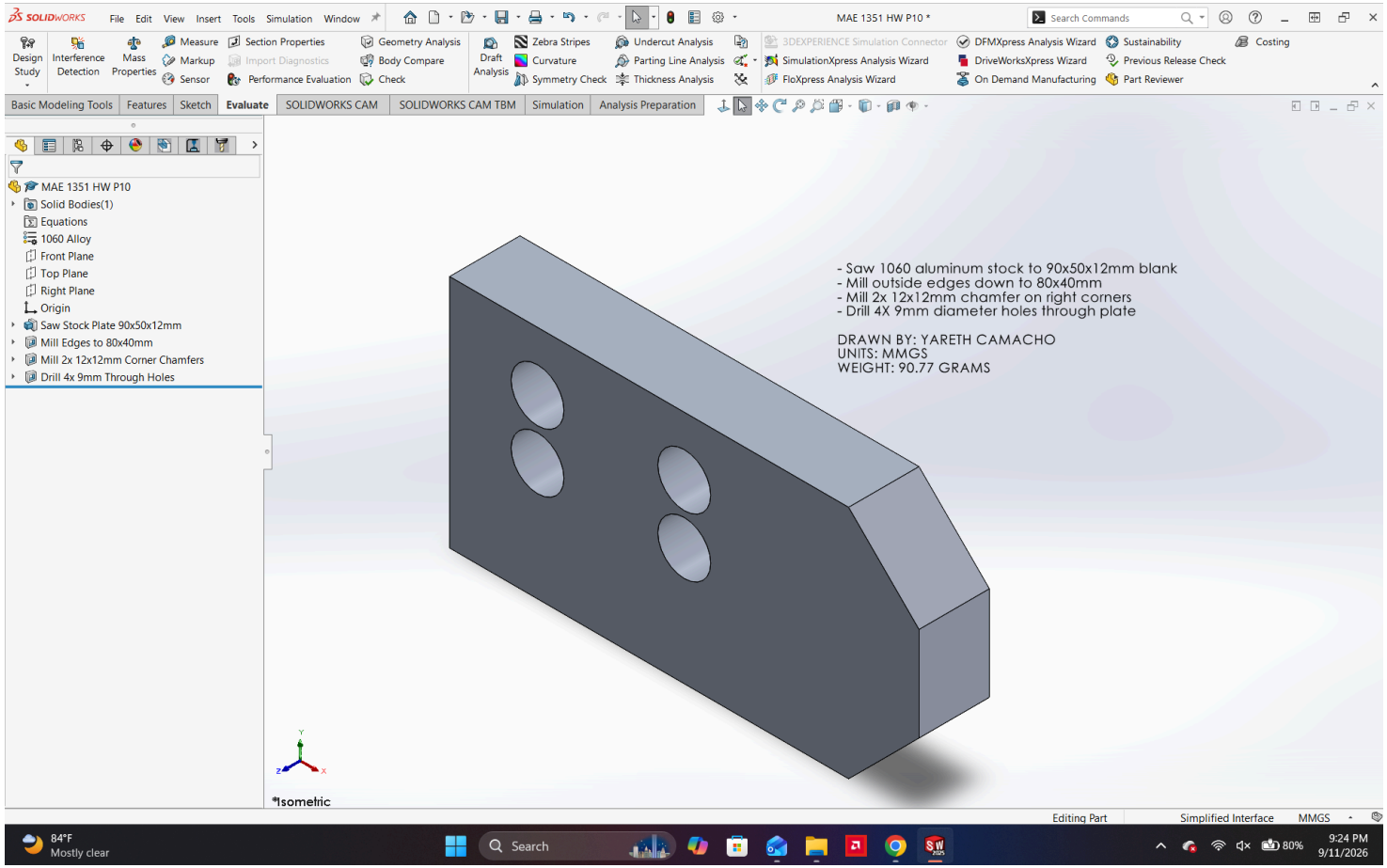
# Question 8


MAE 1351 HW1 P8.2 *
MAE 1351 HW1 P8.2
Solid Bodies(1)
Equations
1060 Alloy
Front Plane
Top Plane
Right Plane
Origin
Revolve1
(-) Sketch3
DRAWN BY: YARETH CAMACHO
UNITS: MMGS
WEIGHTS: 0.21 KG
*Isometric
Editing Part
Simplified Interface
MMGS
80°F
Clear
Search
80%
10:30 PM
9/11/2026

# Question 9


SOLIDWORKS
File
Edit
View
Insert
Tools
Simulation
Window
MAE 1351 HW1 P9 *
Design Study
Interference Detection
Mass Properties
Measure
Markup
Sensor
Section Properties
Import Diagnostics
Performance Evaluation
Geometry Analysis
Body Compare
Check
Draft Analysis
Zebra Stripes
Curvature
Symmetry Check
Undercut Analysis
Parting Line Analysis
Thickness Analysis
3DEXPERIENCE Simulation Connector
SimulationXpress Analysis Wizard
FloXpress Analysis Wizard
DFMXpress Analysis Wizard
DriveWorksXpress Wizard
On Demand Manufacturing
Sustainability
Previous Release Check
Part Reviewer
Costing
Basic Modeling Tools
Features
Sketch
Evaluate
SOLIDWORKS CAM
SOLIDWORKS CAM TBM
Simulation
Analysis Preparation
MAE 1351 HW1 P9
Solid Bodies(1)
Equations
Copper
Front Plane
Top Plane
Right Plane
Origin
Plane1
Sweep-Thin2
Boss-Extrude1
Cut-Extrude1
Boss-Extrude2
Cut-Extrude2
DRAWN BY: YARETH CAMACHO
UNITS: MMGS
WEIGHT: 1.83 KG
*Isometric
SOLIDWORKS Student Edition - Academic Use Only
Editing Part
Simplified Interface
MMGS
84°F
Mostly clear
Search
64%
8:27 PM
9/11/2026

# Question 10


MAE 1351 HW P10
Solid Bodies(1)
Equations
1060 Alloy
Front Plane
Top Plane
Right Plane
Origin
Saw Stock Plate 90x50x12mm
Mill Edges to 80x40mm
Mill 2x 12x12mm Corner Chamfers
Drill 4x 9mm Through Holes
- Saw 1060 aluminum stock to 90x50x12mm blank
- Mill outside edges down to 80x40mm
- Mill 2x 12x12mm chamfer on right corners
- Drill 4X 9mm diameter holes through plate
DRAWN BY: YARETH CAMACHO
UNITS: MMGS
WEIGHT: 90.77 GRAMS
*Isometric